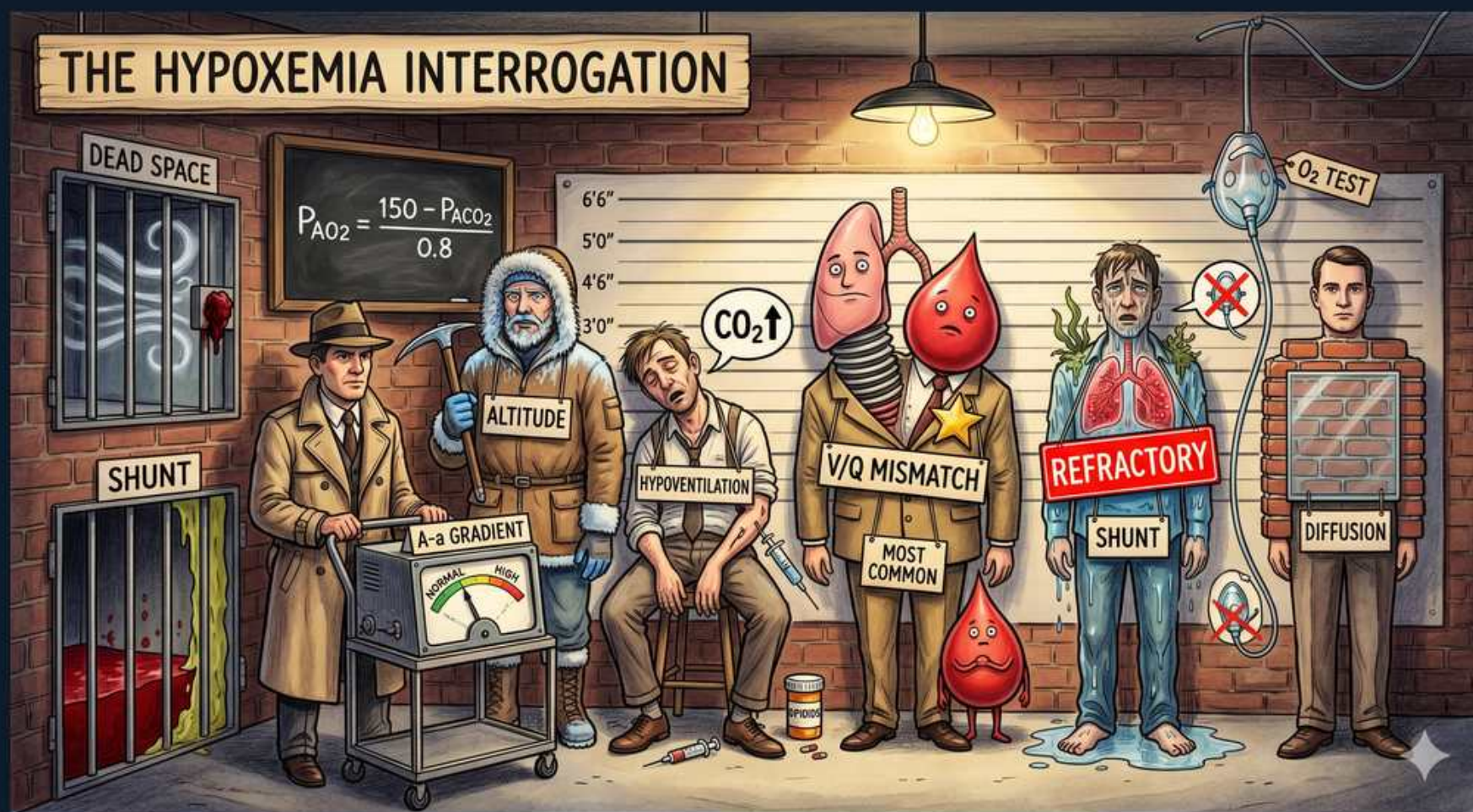


Hypoxemia: 5 Mechanisms, A-a Gradient & Oxygen Response



1. Alveolar gas equation

WHAT YOU SEE

Chalkboard 'PAO2 = 150 - PaCO2/0.8'

WHAT IT ENCODES

Alveolar gas equation: $PAO_2 = FiO_2 \times (P_{atm} - P_{H_2O}) - PaCO_2/R \approx 150 - PaCO_2/0.8$ at sea level on room air. It predicts alveolar O2, the reference for calculating the A-a gradient.

BOARD TRAP

Rising PaCO2 (hypoventilation) lowers PAO2 mathematically — so hypoventilation causes hypoxemia with a NORMAL A-a gradient.

2. A-a gradient gauge

WHAT YOU SEE

'A-a GRADIENT' meter on cart

WHAT IT ENCODES

A-a gradient = PAO2 (calculated) - PaO2 (measured). Normal $\approx 5-15$ (rises with age: $\sim \text{age}/4 + 4$). A normal gradient localizes the problem to low inspired O2 or hypoventilation; a widened gradient indicates lung pathology.

BOARD TRAP

A normal A-a gradient narrows the cause to hypoventilation or high altitude — the other three mechanisms all WIDEN the gradient.

3. Low inspired O2 / altitude

WHAT YOU SEE

Mountaineer labeled 'ALTITUDE'

WHAT IT ENCODES

High altitude/low FiO2 lowers PiO2 and thus PAO2, causing hypoxemia with a NORMAL A-a gradient. Corrects readily with supplemental O2. Triggers hypoxic ventilatory drive → respiratory alkalosis.

BOARD TRAP

Altitude hypoxemia has a normal A-a gradient — same category as hypoventilation, distinct from V/Q, shunt, and diffusion.

4. Hypoventilation

WHAT YOU SEE

Sleepy slumped figure 'HYPOVENTILATION'

WHAT IT ENCODES

Hypoventilation raises PaCO2, which lowers PAO2 and PaO2, giving hypoxemia with a NORMAL A-a gradient that corrects with O2. Causes: CNS depression (opioids), neuromuscular disease, obesity hypoventilation, chest wall disorders.

BOARD TRAP

Hypoventilation is the only hypoxemia mechanism that always RAISES PaCO2 — high CO2 with normal A-a gradient points straight to it.

5. V/Q mismatch (most common)

WHAT YOU SEE

Lungs sheriff, 'V/Q MISMATCH', 'MOST COMMON'

WHAT IT ENCODES

V/Q mismatch is the MOST COMMON cause of hypoxemia (COPD, asthma, PE, pneumonia, interstitial disease). Widened A-a gradient; hypoxemia IMPROVES with supplemental O2 because some perfused alveoli still ventilate.

BOARD TRAP

V/Q mismatch corrects with O2, distinguishing it from true shunt which does NOT correct — the oxygen-response test separates them.

6. CO2 rise in V/Q

WHAT YOU SEE

'CO2↑' speech bubble near lungs

WHAT IT ENCODES

Severe/widespread V/Q mismatch can eventually raise CO2 when ventilation cannot compensate, but early on CO2 is normal/low due to compensatory hyperventilation of healthy units.

BOARD TRAP

Early V/Q mismatch often shows LOW CO2 (hyperventilation) — a normal or high CO2 in a hypoxic COPD patient signals decompensation.

7. Shunt (refractory)

WHAT YOU SEE

Star-marked figure 'SHUNT', 'REFRACTORY' red sign

WHAT IT ENCODES

True shunt: blood bypasses ventilated alveoli (atelectasis, pneumonia consolidation, ARDS, pulmonary AVM, intracardiac R→L shunt). Wide A-a gradient and hypoxemia REFRACTORY to 100% O2 — the defining feature.

BOARD TRAP

Failure to correct with 100% O2 = shunt, not V/Q mismatch — this single test is the classic board discriminator.

8. Shunt does not correct with O2

WHAT YOU SEE

'REFRACTORY' banner over shunt figure

WHAT IT ENCODES

In shunt, supplemental O2 cannot reach the bypassed blood, so PaO2 stays low despite high FiO2. Treat the underlying lung collapse/consolidation (PEEP, recruitment) rather than just increasing FiO2.

BOARD TRAP

A shunt fraction >30-50% makes oxygen therapy nearly futile — escalate to PEEP/mechanical recruitment, not just more O2.

9. Diffusion limitation

WHAT YOU SEE

Right figure behind mesh 'DIFFUSION'

WHAT IT ENCODES

Diffusion impairment (interstitial lung disease, emphysema, pulmonary fibrosis) widens the A-a gradient; hypoxemia worsens with exercise and improves with supplemental O₂. Reduced DLCO on PFTs.

BOARD TRAP

Diffusion-limited hypoxemia is often only apparent on EXERTION — resting ABG can be normal in early ILD; check exercise desaturation/DLCO.

10. O₂ test (response to oxygen)

WHAT YOU SEE

'O₂ TEST' tag, hanging O₂ mask

WHAT IT ENCODES

The 100% O₂ challenge classifies hypoxemia: corrects in low-FiO₂, hypoventilation, V/Q mismatch, and diffusion; does NOT fully correct in shunt. This is the key bedside discriminator.

BOARD TRAP

Only TRUE SHUNT fails to correct with 100% O₂ — all four other mechanisms improve, making the O₂ test diagnostic.

11. Dead space (jail cell)

WHAT YOU SEE

Left cell labeled 'DEAD SPACE'

WHAT IT ENCODES

Dead space = ventilation without perfusion ($V/Q \rightarrow \infty$), classically pulmonary embolism. Wasted ventilation raises required minute ventilation; in pure dead space, gas exchange units are ventilated but not perfused.

BOARD TRAP

PE causes BOTH dead space (occluded segments) and hypoxemia via V/Q mismatch/shunt in surrounding lung — not pure dead space alone.

12. Shunt (jail cell)

WHAT YOU SEE

Left lower cell labeled 'SHUNT'

WHAT IT ENCODES

Shunt = perfusion without ventilation ($V/Q \rightarrow 0$), the opposite extreme of dead space. Blood passes unventilated alveoli (atelectasis, ARDS) and stays deoxygenated, producing refractory hypoxemia.

BOARD TRAP

Dead space ($V/Q=\infty$) and shunt ($V/Q=0$) are the two V/Q EXTREMES — confusing which has perfusion vs ventilation is a common error.

13. V/Q extremes spectrum

WHAT YOU SEE

Lineup arrangement spanning dead space to shunt

WHAT IT ENCODES

The lineup represents the V/Q continuum: dead space (high V/Q, no perfusion) on one end, shunt (low V/Q, no ventilation) on the other, with normal matching in between. Most disease lies along this spectrum.

BOARD TRAP

Pure shunt and pure dead space are extremes; most real disease is intermediate V/Q mismatch — the commonest hypoxemia mechanism overall.

14. Hypoxemia vs hypoxia

WHAT YOU SEE

Title 'THE HYPOXEMIA INTERROGATION'

WHAT IT ENCODES

Hypoxemia = low arterial PaO₂/SaO₂ (a blood measurement); hypoxia = inadequate tissue O₂ delivery. Hypoxia can occur with normal PaO₂ (anemia, CO poisoning, cyanide, shock).

BOARD TRAP

CO poisoning and anemia cause tissue HYPOXIA with a NORMAL PaO₂ and normal pulse-ox — measure co-oximetry, not just SpO₂.